AI23331 Fundamentals of Machine Learning

HOUSE PRICE PREDICTION AND ANALYSIS

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report "HOUSE PRICE PREDICTION AND ANALYSIS" is the bonafide work of "DHAANYA G K (230701069) , JANANIE (230701124) " who carried out the project work under my supervision.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	04
1	INTRODUCTION	05
2	LITERATURE REVIEW	05
3	PROPOSED SYSTEM	07
	3.1 ARCHITECTURE DIAGRAM	08
4	MODULES DESCRIPTION	
	4.1 MODULE 1	09
	4.2 MODULE 2	09
	4.3 MODULE 3	10
	4.4 MODULE 4	10
5	IMPLEMENTATION AND RESULTS	
	5.1 EXPERIMENTAL SETUP	11
	5.2 RESULTS	12
6	CONCLUSION AND FUTURE WORK	13
7	REFERENCES	14

ABSTRACT

This project aims to develop a machine learning-based system for predicting house prices using important features such as area, number of bedrooms, and other property-related attributes. Accurate price prediction is essential in the real estate sector to assist buyers, sellers, and investors in making informed decisions. The dataset is preprocessed by handling missing values, selecting relevant features, and performing exploratory data analysis to understand the relationship between variables. Multiple machine learning algorithms, including Linear Regression, Decision Tree Regression, and Random Forest Regression, are implemented and compared to achieve better prediction performance. Among these models, Random Forest Regression provides the most accurate results due to its ability to capture complex patterns and reduce overfitting. The models are evaluated using performance metrics such as Mean Absolute Error (MAE) and R-squared score. A user-friendly web interface is developed using Streamlit, enabling users to input property details and obtain instant predictions. In conclusion, the project demonstrates that machine learning techniques can effectively predict house prices and provide a reliable, fast, and data-driven solution for real estate price estimation.

1. INTRODUCTION

The real estate sector is a rapidly growing industry where accurate estimation of property prices is essential for buyers, sellers, and investors. Traditional methods rely on manual analysis and market trends, which can be time-consuming and less accurate. With the advancement of machine learning, it is possible to analyze large housing datasets and identify complex patterns that influence property prices. This project focuses on developing a machine learning-based system to predict house prices using features such as area, number of bedrooms, location, and other relevant factors. Data preprocessing techniques, including handling missing values and feature selection, are applied to improve data quality. Exploratory Data Analysis (EDA) is performed to understand relationships between variables. Multiple algorithms such as Linear Regression, Decision Tree Regression, and Random Forest Regression are implemented and compared. Among these, Random Forest provides better accuracy by capturing complex patterns. The models are evaluated using metrics like Mean Absolute Error (MAE) and R-squared score. The system is deployed using a Streamlit-based web application, allowing users to input property details and obtain instant predictions. Overall, the project demonstrates how machine learning provides a fast, accurate, and data-driven solution for house price prediction.

2. LITERATURE REVIEW

House price prediction has been a widely studied problem in the field of machine learning and data analysis. Various researchers have explored different techniques to improve prediction accuracy and understand the factors influencing property prices. Early studies mainly used statistical methods such as Linear Regression to estimate house prices based on features like area, number of rooms, and location [5]. These methods were simple and easy to implement but had limitations in capturing complex and non-linear relationships present in real-world data. With the advancement of machine learning, more sophisticated models such as Decision Tree Regression were introduced [7]. Decision Trees improved prediction performance by splitting data into smaller subsets and handling non-linear relationships more effectively. However, these models were prone to overfitting, especially when trained on large datasets. To overcome these limitations, ensemble learning methods such as Random Forest Regression were proposed [1]. Random Forest combines multiple decision trees to produce more accurate and stable predictions. It reduces overfitting and improves generalization, making it one of the most widely used algorithms for regression problems like house price prediction. Recent studies have also explored modern tools and implementations using machine learning libraries such as Scikit-learn [6] and practical approaches discussed in advanced resources [3][8]. Additionally, feature engineering and data preprocessing play a crucial role in improving model performance. Many datasets used for house price prediction are publicly available on platforms like Kaggle [4], which provide real-world data for training and testing models. Furthermore, deep learning approaches have also been explored to enhance prediction accuracy [2]. This project builds upon these existing approaches by implementing and comparing multiple machine learning algorithms, including Linear Regression, Decision Tree Regression, and Random Forest Regression, along with a user-friendly interface for real-time house price prediction.

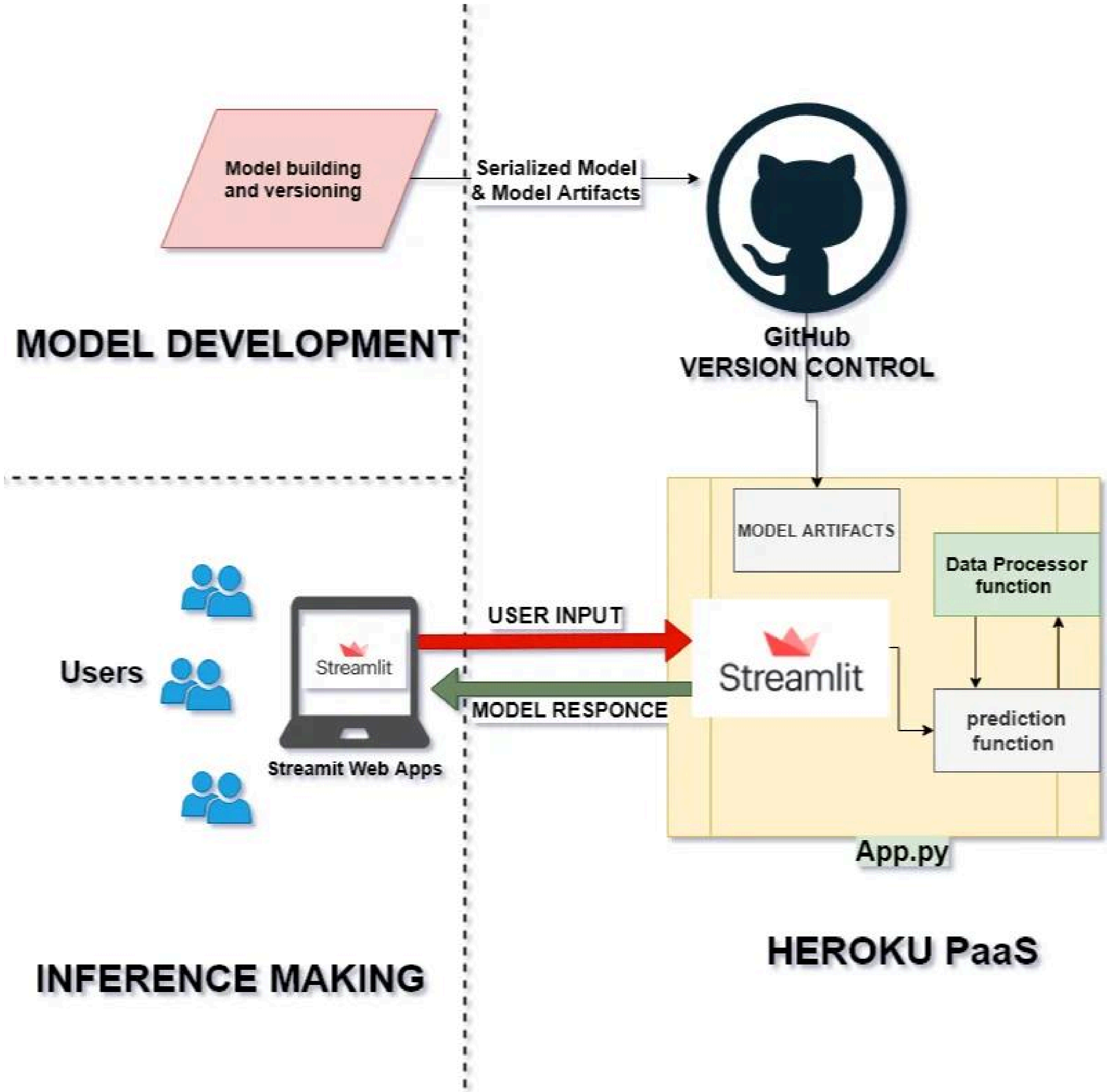
3. PROPOSED SYSTEM

The proposed system is a machine learning-based web application designed to predict house prices accurately using user-provided inputs. The system aims to overcome the limitations of traditional price estimation methods by using data-driven techniques and multiple regression algorithms. The system takes key input features such as area, number of bedrooms, bathrooms, and other relevant parameters from the user. These inputs are processed using a trained machine learning model to generate accurate price predictions. The proposed system follows a structured workflow. Initially, the dataset is collected and preprocessed by handling missing values, removing inconsistencies, and selecting important features. Exploratory Data Analysis (EDA) is performed to understand relationships between variables. The processed data is then used to train multiple models, including Linear Regression, Decision Tree Regression, and Random Forest Regression. Among these, Random Forest Regression is selected as the final model due to its higher accuracy and ability to handle complex patterns while reducing overfitting. The trained model is then deployed using a Streamlit-based web interface, allowing users to interact with the system easily.

The proposed system provides the following advantages:

- Improved prediction accuracy
- Reduced human error
- Fast and real-time predictions
- User-friendly interface
- Scalable and adaptable for future improvements

3.1. ARCHITECTURAL DIAGRAM



4. MODULES DESCRIPTION

4.1. MODULE 1: Data Collection and Preprocessing

This module is responsible for collecting and preparing the dataset for model training. The housing dataset (CSV file) contains features such as price, area, number of bedrooms, and other relevant attributes. In this module, data preprocessing techniques are applied to improve data quality. These include handling missing values, removing duplicate entries, and selecting important features. Additionally, exploratory data analysis (EDA) is performed to understand the relationship between input features and house prices. Proper preprocessing ensures that the data is clean, structured, and suitable for training machine learning models.

4.2. MODULE 2: Model Training and Evaluation

This module focuses on building and evaluating machine learning models for house price prediction. Multiple algorithms such as Linear Regression, Decision Tree Regression, and Random Forest Regression are implemented using the processed dataset. The dataset is divided into training and testing sets to evaluate model performance. Each model is trained and tested using evaluation metrics such as Mean Absolute Error (MAE) and R-squared score. Based on performance comparison, the best model (Random Forest Regression) is selected due to its higher accuracy and ability to handle complex patterns. The final trained model is then saved as a serialized file (.pkl) for later use in prediction.

4.3. MODULE 3: Model Deployment (Streamlit)

This module focuses on deploying the trained machine learning model into a real-time application. The best-performing model, saved as a serialized file (model.pkl), is integrated into a web application using Streamlit. The deployment process involves loading the trained model into the application (app.py) and ensuring that it can process user inputs efficiently. Streamlit is used to create an interactive and lightweight web interface without requiring complex backend development. This module enables real-time predictions by connecting the trained model with the frontend interface. It ensures that the system is accessible, easy to use, and capable of delivering instant results. The deployment makes the model practical for real-world use rather than limiting it to offline analysis.

4.4 Module 4: User Interface and Prediction System

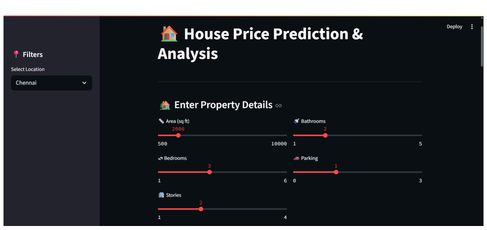
This module handles user interaction and prediction generation. A user-friendly interface is designed using Streamlit, where users can input property details such as area, number of bedrooms, and other relevant features. Once the user provides input, the system processes the data using a preprocessing function to match the format required by the trained model. The processed input is then passed to the prediction function, which uses the machine learning model to estimate the house price. The predicted result is displayed instantly on the interface, providing a seamless user experience. This module ensures smooth communication between the user and the machine learning model, making the system efficient, interactive, and easy to operate.

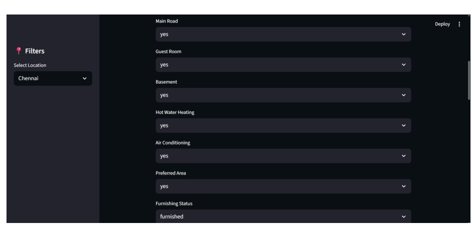
5. IMPLEMENTATION AND RESULTS

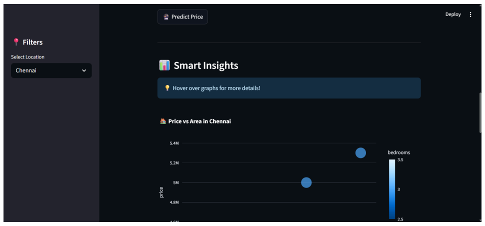
5.1. Experimental Setup

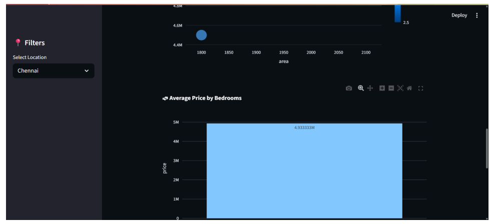
The house price prediction system is implemented using Python and various machine learning libraries. The development is carried out in Visual Studio Code, and the application is deployed using Streamlit for a user-friendly interface. The dataset used for this project is a housing dataset in CSV format containing features such as price, area, number of bedrooms, and other relevant attributes. The data is preprocessed by handling missing values, removing inconsistencies, and selecting important features to improve model performance. The implementation involves training multiple machine learning algorithms, including Linear Regression, Decision Tree Regression, and Random Forest Regression, using the Scikit-learn library. The dataset is split into training and testing sets to evaluate the model performance. Standard evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared score are used. The final model is selected based on its performance and is saved as a serialized file (model.pkl). This model is then integrated into a Streamlit-based web application (app.py), which allows users to input property details and obtain predictions in real time.

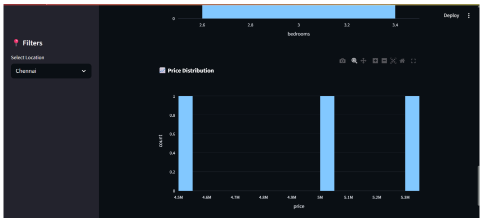
OUTPUT











5.2. Results

The performance of different machine learning models is compared based on evaluation metrics. Among the implemented models, Random Forest Regression provides the best results with higher accuracy and lower error values compared to Linear Regression and Decision Tree Regression. The model achieves a high R-squared score (approximately around 0.85–0.90), indicating that it can effectively predict house prices based on input features. The Mean Absolute Error (MAE) is relatively low, showing that the predicted prices are close to the actual values. The system successfully generates real-time predictions through the Streamlit interface. Users can enter property details such as area and number of bedrooms, and the system instantly displays the predicted house price. Overall, the results demonstrate that the proposed system is efficient, accurate, and practical for real-world applications. The use of ensemble methods like Random Forest significantly improves prediction performance and reliability.

6. CONCLUSION AND FUTURE WORK

6.1. Conclusion

This project successfully demonstrates the use of machine learning techniques for predicting house prices based on various features such as area, number of bedrooms, and other relevant attributes. Multiple models, including Linear Regression, Decision Tree Regression, and Random Forest Regression, were implemented and compared to identify the most effective approach. Among these, Random Forest Regression produced the best results due to its ability to handle complex relationships and reduce overfitting. The model achieved good accuracy with a high R-squared score and low error values, making it reliable for prediction tasks. The system was further enhanced by deploying the model using a Streamlit-based web application, allowing users to input property details and obtain instant predictions. This makes the solution practical, user-friendly, and suitable for real-world applications. Overall, the project highlights the effectiveness of machine learning in providing fast, accurate, and data-driven solutions in the real estate domain.

6.2. Future Work

Incorporating additional features such as location coordinates, nearby facilities, and market trends to improve prediction accuracy. Using advanced machine learning algorithms like Gradient Boosting or XGBoost for better performance. Integrating real-time data from online real estate platforms.Enhancing the user interface with better visualization and interactive features. Deploying the application on cloud platforms for wider accessibility. Adding a recommendation system to suggest suitable properties based on user preferences.

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